KARTHIKEYAN KUMARAGURUPARAN

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SUMMARY

Software engineer specializing in backend and distributed systems. Year-long software engineering internship building production multi-tenant microservices, durable Temporal workflows, and serverless AWS onboarding; now completing an MS in Computer Science at USC (May 2027). Systems projects focus on failure behavior — quorum-replicated storage, crash-safe LLM-agent execution, and a Linux energy profiler — all open source on GitHub with green CI.

EDUCATION

University of Southern California **Aug 2025 – May 2027**
Master of Science in Computer Science — GPA: 3.53/4.00 *Los Angeles, CA*
Relevant Coursework: Distributed Systems, Algorithms & Data Structures, Database Management

Kumaraguru College of Technology **Nov 2020 – May 2024**
Bachelor of Engineering in Computer Science and Engineering — GPA: 9.23/10.0 *Coimbatore, India*

TECHNICAL SKILLS

Languages: Go, Python, C++, Java, TypeScript, JavaScript, SQL, Bash
Backend & APIs: Node.js, Express.js, REST APIs, gRPC, Protocol Buffers, Temporal, Microservices
Cloud & DevOps: AWS (Lambda, API Gateway, IAM, KMS), Docker, Jenkins, GitHub Actions, Ansible, Nginx, Linux, Git
Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, RocksDB, SQLite
Distributed Systems: Quorum replication, write-ahead logging (WAL), crash recovery, idempotency, fault injection, linearizability checking

PROFESSIONAL EXPERIENCE

Clar Technologies — Client: Skuchain **Jul 2023 – Jul 2024**
Software Engineer Intern · Node.js, Temporal, AWS, Docker, Ansible, MongoDB *Coimbatore, India*

- Built Node.js/Express microservices for a multi-tenant supply-chain-financing platform, running long multi-step financing operations as durable Temporal workflows that survive restarts and partial failures.
- Built serverless onboarding and key-management flows on AWS Lambda (API Gateway, IAM, KMS) serving 500+ users; documented 8+ partner REST APIs for external integrators.
- Cut environment provisioning from ~60 minutes to ~5 with Ansible playbooks; containerized services with Docker and standardized Jenkins delivery pipelines across a distributed team.
- Operated the platform's Nginx reverse proxy for 12 months without a recorded outage.

Samsung PRISM — Samsung R&D Institute India **Mar 2023 – Sep 2023**
Project Intern — raplscope: Linux System Energy Profiler · Go, Linux *Remote*

- Created raplscope, a zero-dependency Go CLI measuring whole-system energy and power from Linux RAPL counters via powercap sysfs, sampling down to a validated 10 ms floor; since rebuilt end to end and published at github.com/KarthikeyanK1206/raplscope.
- Engineered wraparound-safe accounting: wrap-corrected deltas (counters wrap roughly every 45 min at ~100 W), boundary snapshots so sub-interval commands measure correctly, monotonic-clock power math, and package-only summation to avoid double counting.
- Added /usr/bin/time-style command wrapping — inherited streams, SIGINT/SIGTERM forwarding, exit-code propagation (128+N on signals) — with JSON/CSV export feeding a bundled HTML energy dashboard.
- Made hardware-bound logic testable without root or RAPL hardware by injecting the sysfs root; 12 tests — table-driven wrap math plus discovery/read tests against a fake counter tree.

PROJECTS

NimbusVault — Adaptive Distributed Object Store **C++17, gRPC, Protocol Buffers, RocksDB · Group Project**

- Built the metadata plane: a RocksDB-backed write-ahead log replicated from a fixed-leader coordinator across a 3-node cluster with quorum commits — deliberately scoped below full Raft (no election or membership changes).
- Designed adaptive replication: access windows classify chunks hot/warm/cold with asymmetric hysteresis (3 windows to promote, 5 to demote), scaling between 2 and 5 replicas — cold data holds 2 replicas vs. a static 3, a one-third cold-storage reduction by construction.
- Built two-stage replica reconfiguration: a quorum-committed START keeps the old set readable while new copies are made and verified, then COMMIT publishes the stable set; a Wing–Gong checker accepted 60/60 keys in each recorded 6,060-op concurrent history, including during these transitions (single-machine runs).
- Validated with 83/83 unit tests and 10/10 live-cluster integration scenarios in GitHub Actions CI, covering storage-node kills, metadata follower catch-up, corrupt-replica detection and bypass, and input validation.

RAEF — Fault-Tolerant Execution Runtime for LLM Agents **Python, SQLite, LangChain, OpenRouter · Group Project**

- Designed a log-before-send transaction layer: every tool call is journaled to SQLite (WAL, synchronous=FULL) before dispatch under a deterministic SHA-256-derived execution ID, so a restarted agent reuses committed results instead of re-firing writes.
- Built verification-driven recovery: an ambiguous write parks in a durable pending state until a pluggable verifier proves it committed (reuse) or absent (replay exactly once), else hands off; an operator CLI audits any run's log-first invariants.
- Recorded a nine-phase crash-injection matrix: every phase ends with exactly one mock-target commit and zero duplicates, asserted from the target store's own commit counter — 47-test suite, zero-dependency core, CI green on Python 3.12/3.13.

PUBLICATION

S. Barath Sanjay, B. Harish, K. Karthikeyan, et al., "IoT Based Smart IV Drip Stand," *IEEE Conference*, Oct. 2021.